

The FREQ Procedure

Table of coffee by MI			
coffee	MI		
	0	1	Total
0	380	220	600
	31.67	18.33	50.00
	63.33	36.67	
	63.33	36.67	
1	220	380	600
	18.33	31.67	50.00
	36.67	63.33	
	36.67	63.33	
Total	600	600	1200
	50.00	50.00	100.00

Statistics for Table of coffee by MI

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	2.9835	2.3591	3.7731
Relative Risk (Column 1)	1.7273	1.5296	1.9504
Relative Risk (Column 2)	0.5789	0.5127	0.6537

Sample Size = 1200

The FREQ Procedure

Summary Statistics for coffee by MI

Cochran-Mantel-Haenszel Statistics (Based on Table Scores)				
Statistic	Alternative Hypothesis	DF	Value	Prob
1	Nonzero Correlation	1	85.2622	<.0001
2	Row Mean Scores Differ	1	85.2622	<.0001
3	General Association	1	85.2622	<.0001

Common Odds Ratio and Relative Risks				
Statistic	Method	Value	95% Confidence Limits	
Odds Ratio	Mantel-Haenszel	2.9835	2.3591	3.7731
	Logit	2.9835	2.3591	3.7731
Relative Risk (Column 1)	Mantel-Haenszel	1.7273	1.5296	1.9504
	Logit	1.7273	1.5296	1.9504
Relative Risk (Column 2)	Mantel-Haenszel	0.5789	0.5127	0.6537
	Logit	0.5789	0.5127	0.6537

Total Sample Size = 1200

Q1c: Smoking-adjusted Coffee?MI Association (Mantel?Haenszel + Zelen test)

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table 1 of coffee by MI			
	Controlling for smoking=0			
	coffee	MI		
		0	1	Total
0	303 50.50 73.01 73.01	112 18.67 26.99 60.54	415 69.17	
1	112 18.67 60.54 26.99	73 12.17 39.46 39.46	185 30.83	
Total	415 69.17	185 30.83	600 100.00	

Statistics for Table 1 of coffee by MI
Controlling for smoking=0

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	1.7633	1.2230	2.5424
Relative Risk (Column 1)	1.2060	1.0588	1.3737
Relative Risk (Column 2)	0.6839	0.5388	0.8682

Sample Size = 600

Frequency Percent Row Pct Col Pct	Table 2 of coffee by MI			
	Controlling for smoking=1			
	coffee	MI		
		0	1	Total
		0	77 12.83 41.62 41.62	108 18.00 58.38 26.02
1	108 18.00 26.02 58.38	307 51.17 73.98 73.98	415 69.17	
Total	185 30.83	415 69.17	600 100.00	

Statistics for Table 2 of coffee by MI
Controlling for smoking=1

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	2.0267	1.4063	2.9207
Relative Risk (Column 1)	1.5993	1.2638	2.0239
Relative Risk (Column 2)	0.7892	0.6899	0.9027

Sample Size = 600

Q1c: Smoking-adjusted Coffee?MI Association (Mantel-Haenszel + Zelen test)

The FREQ Procedure

Summary Statistics for coffee by MI
Controlling for smoking

Cochran-Mantel-Haenszel Statistics (Based on Table Scores)				
Statistic	Alternative Hypothesis	DF	Value	Prob
1	Nonzero Correlation	1	23.5967	<.0001
2	Row Mean Scores Differ	1	23.5967	<.0001
3	General Association	1	23.5967	<.0001

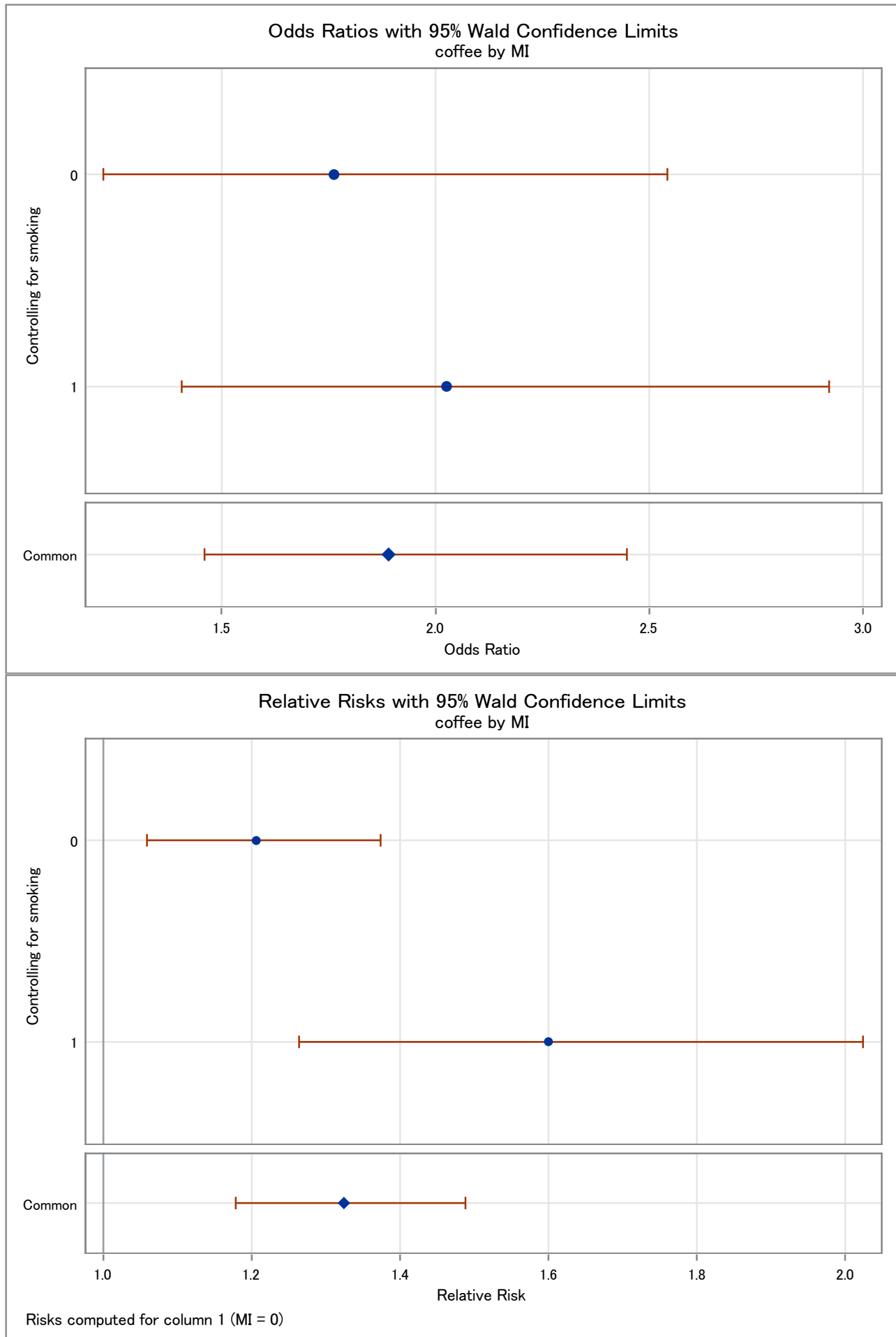
Common Odds Ratio and Relative Risks				
Statistic	Method	Value	95% Confidence Limits	
Odds Ratio	Mantel-Haenszel	1.8902	1.4597	2.4477
	Logit	1.8906	1.4598	2.4485
Relative Risk (Column 1)	Mantel-Haenszel	1.3243	1.1785	1.4881
	Logit	1.2884	1.1497	1.4439
Relative Risk (Column 2)	Mantel-Haenszel	0.7526	0.6677	0.8482
	Logit	0.7624	0.6782	0.8571

Tests for Homogeneity of Odds Ratios	
Breslow-Day Chi-Square	0.2784
DF	1
Pr > ChiSq	0.5978
Zelen's Exact Test (P)	0.0914
Exact Pr <= P	0.6921

Total Sample Size = 1200

Q1c: Smoking-adjusted Coffee?MI Association (Mantel-Haenszel + Zelen test)

The FREQ Procedure



The LOGISTIC Procedure

Model Information	
Data Set	WORK.MI
Response Variable	MI
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1200
Number of Observations Used	1200

Response Profile		
Ordered Value	MI	Total Frequency
1	1	600
2	0	600

Probability modeled is MI=1.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1665.553	1581.179
SC	1670.643	1591.359
-2 Log L	1663.553	1577.179

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	86.3746	1	<.0001
Score	85.3333	1	<.0001
Wald	83.2341	1	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-0.5465	0.0847	41.6170	<.0001
coffee	1	1.0930	0.1198	83.2341	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
coffee	2.983	2.359	3.773

The LOGISTIC Procedure

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	40.1	Somers' D	0.267
Percent Discordant	13.4	Gamma	0.498
Percent Tied	46.4	Tau-a	0.133
Pairs	360000	c	0.633

The LOGISTIC Procedure

Model Information	
Data Set	WORK.MI
Response Variable	MI
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1200
Number of Observations Used	1200

Response Profile		
Ordered Value	MI	Total Frequency
1	1	600
2	0	600

Probability modeled is MI=1.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1665.553	1467.278
SC	1670.643	1487.638
-2 Log L	1663.553	1459.278

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	204.2755	3	<.0001
Score	196.7463	3	<.0001
Wald	181.4267	3	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-0.9952	0.1106	80.9958	<.0001
smoking	1	1.3336	0.1857	51.5844	<.0001
coffee	1	0.5672	0.1867	9.2296	0.0024
smoking*coffee	1	0.1392	0.2639	0.2783	0.5978

The LOGISTIC Procedure

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	60.6	Somers' D	0.443
Percent Discordant	16.2	Gamma	0.577
Percent Tied	23.2	Tau-a	0.222
Pairs	360000	c	0.722

Q1c: Logistic Regression ? No Interaction (adjusted)

The LOGISTIC Procedure

Model Information	
Data Set	WORK.MI
Response Variable	MI
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	1200
Number of Observations Used	1200

Response Profile		
Ordered Value	MI	Total Frequency
1	1	600
2	0	600

Probability modeled is MI=1.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	1665.553	1465.556
SC	1670.643	1480.826
-2 Log L	1663.553	1459.556

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	203.9971	2	<.0001
Score	196.4963	2	<.0001
Wald	181.2457	2	<.0001

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-1.0198	0.1009	102.2586	<.0001
smoking	1	1.4028	0.1319	113.1185	<.0001
coffee	1	0.6368	0.1319	23.3124	<.0001

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
smoking	4.067	3.140	5.266
coffee	1.890	1.460	2.448

Q1c: Logistic Regression ? No Interaction (adjusted)

The LOGISTIC Procedure

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	60.6	Somers' D	0.443
Percent Discordant	16.2	Gamma	0.577
Percent Tied	23.2	Tau-a	0.222
Pairs	360000	c	0.722

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table of smoke by asthma			
	smoke	asthma		
		0	1	Total
	0	576 73.85 90.42 83.00	61 7.82 9.58 70.93	637 81.67
	1	118 15.13 82.52 17.00	25 3.21 17.48 29.07	143 18.33
	Total	694 88.97	86 11.03	780 100.00

Statistics for Table of smoke by asthma

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	2.0006	1.2064	3.3176
Relative Risk (Column 1)	1.0958	1.0120	1.1866
Relative Risk (Column 2)	0.5478	0.3568	0.8409

Sample Size = 780

The FREQ Procedure

Summary Statistics for smoke by asthma

Cochran-Mantel-Haenszel Statistics (Based on Table Scores)				
Statistic	Alternative Hypothesis	DF	Value	Prob
1	Nonzero Correlation	1	7.4321	0.0064
2	Row Mean Scores Differ	1	7.4321	0.0064
3	General Association	1	7.4321	0.0064

Common Odds Ratio and Relative Risks				
Statistic	Method	Value	95% Confidence Limits	
Odds Ratio	Mantel-Haenszel	2.0006	1.2064	3.3176
	Logit	2.0006	1.2064	3.3176
Relative Risk (Column 1)	Mantel-Haenszel	1.0958	1.0120	1.1866
	Logit	1.0958	1.0120	1.1866
Relative Risk (Column 2)	Mantel-Haenszel	0.5478	0.3568	0.8409
	Logit	0.5478	0.3568	0.8409

Total Sample Size = 780

Q2b/Q2c: Sex-specific Smoking?Asthma ORs (Breslow-Day + Zelen test)

The FREQ Procedure

Frequency Percent Row Pct Col Pct	Table 1 of smoke by asthma			
	Controlling for sex=0			
	smoke	asthma		
		0	1	Total
0	315 72.25 88.48 83.33	41 9.40 11.52 70.69	356 81.65	
1	63 14.45 78.75 16.67	17 3.90 21.25 29.31	80 18.35	
Total	378 86.70	58 13.30	436 100.00	

Statistics for Table 1 of smoke by asthma
Controlling for sex=0

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	2.0732	1.1077	3.8800
Relative Risk (Column 1)	1.1236	0.9967	1.2666
Relative Risk (Column 2)	0.5420	0.3252	0.9032

Sample Size = 436

Frequency Percent Row Pct Col Pct	Table 2 of smoke by asthma			
	Controlling for sex=1			
	smoke	asthma		
		0	1	Total
0	261 75.87 92.88 82.59	20 5.81 7.12 71.43	281 81.69	
1	55 15.99 87.30 17.41	8 2.33 12.70 28.57	63 18.31	
Total	316 91.86	28 8.14	344 100.00	

Statistics for Table 2 of smoke by asthma
Controlling for sex=1

Odds Ratio and Relative Risks			
Statistic	Value	95% Confidence Limits	
Odds Ratio	1.8982	0.7953	4.5306
Relative Risk (Column 1)	1.0639	0.9631	1.1753
Relative Risk (Column 2)	0.5605	0.2587	1.2142

Sample Size = 344

Q2b/Q2c: Sex-specific Smoking?Asthma ORs (Breslow-Day + Zelen test)

The FREQ Procedure

Summary Statistics for smoke by asthma
Controlling for sex

Cochran-Mantel-Haenszel Statistics (Based on Table Scores)				
Statistic	Alternative Hypothesis	DF	Value	Prob
1	Nonzero Correlation	1	7.4664	0.0063
2	Row Mean Scores Differ	1	7.4664	0.0063
3	General Association	1	7.4664	0.0063

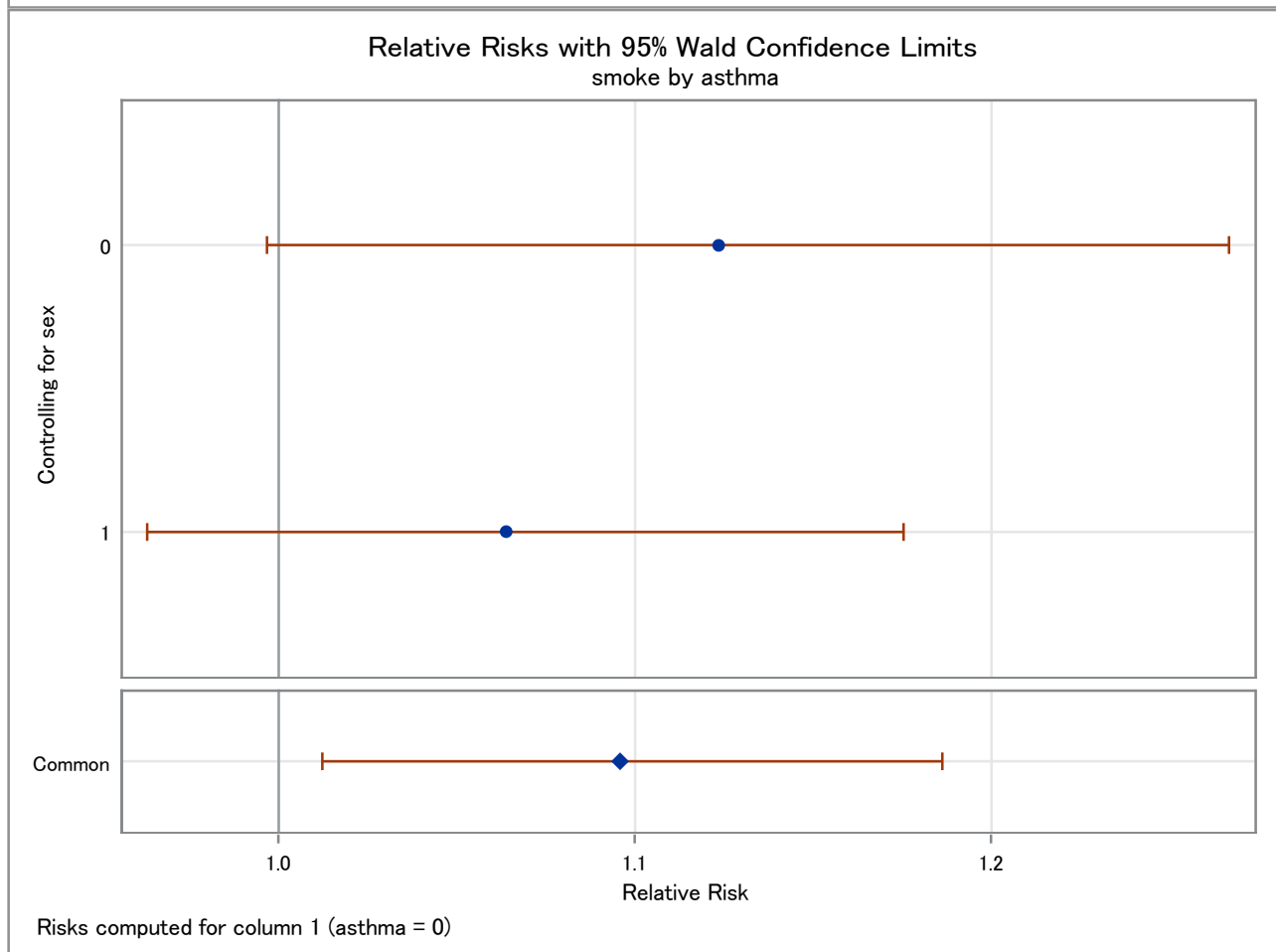
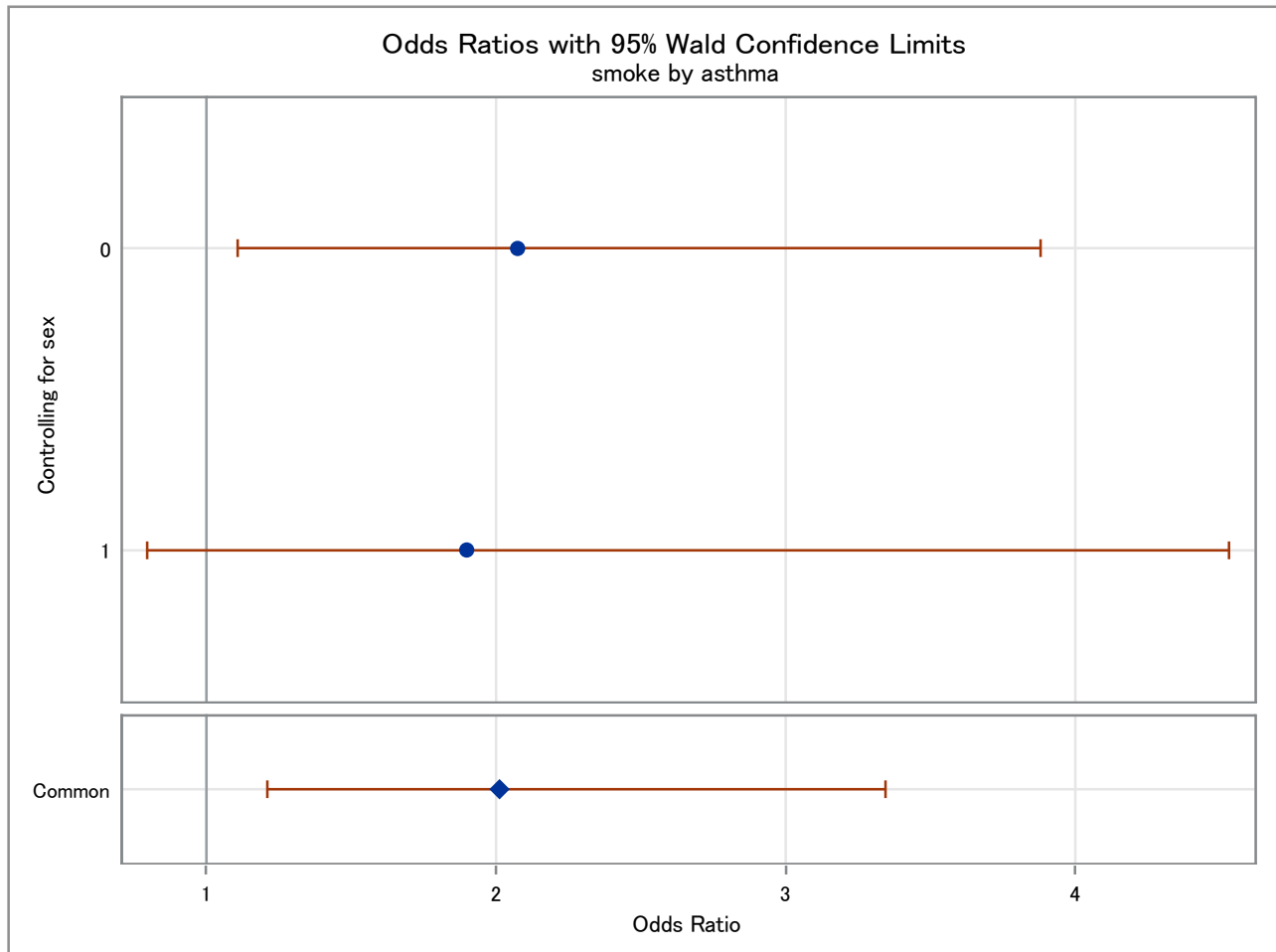
Common Odds Ratio and Relative Risks				
Statistic	Method	Value	95% Confidence Limits	
Odds Ratio	Mantel-Haenszel	2.0118	1.2102	3.3446
	Logit	2.0116	1.2098	3.3450
Relative Risk (Column 1)	Mantel-Haenszel	1.0958	1.0122	1.1862
	Logit	1.0879	1.0077	1.1745
Relative Risk (Column 2)	Mantel-Haenszel	0.5479	0.3577	0.8393
	Logit	0.5475	0.3576	0.8385

Tests for Homogeneity of Odds Ratios	
Breslow-Day Chi-Square	0.0260
DF	1
Pr > ChiSq	0.8719
Zelen's Exact Test (P)	0.2128
Exact Pr <= P	1.0000

Total Sample Size = 780

Q2b/Q2c: Sex-specific Smoking?Asthma ORs (Breslow-Day + Zelen test)

The FREQ Procedure



Q2d: Logistic Regression ? Adjusted, No Interaction

The LOGISTIC Procedure

Model Information	
Data Set	WORK.ASTHMA
Response Variable	asthma
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	780
Number of Observations Used	780

Response Profile		
Ordered Value	asthma	Total Frequency
1	1	86
2	0	694

Probability modeled is asthma=1.

Model Convergence Status	
Convergence criterion (GCONV=1E-8) satisfied.	

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	543.400	535.307
SC	548.059	549.285
-2 Log L	541.400	529.307

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	12.0929	2	0.0024
Score	12.6615	2	0.0018
Wald	12.2253	2	0.0022

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-2.0309	0.1577	165.8936	<.0001
smoke	1	0.6987	0.2593	7.2612	0.0070
sex	1	-0.5540	0.2435	5.1746	0.0229

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
smoke	2.011	1.210	3.343
sex	0.575	0.357	0.926

Q2d: Logistic Regression ? Adjusted, No Interaction

The LOGISTIC Procedure

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	43.6	Somers' D	0.202
Percent Discordant	23.5	Gamma	0.301
Percent Tied	32.9	Tau-a	0.040
Pairs	59684	c	0.601

The LOGISTIC Procedure

Model Information	
Data Set	WORK.ASTHMA
Response Variable	asthma
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	780
Number of Observations Used	780

Response Profile		
Ordered Value	asthma	Total Frequency
1	1	86
2	0	694

Probability modeled is asthma=1.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	543.400	537.281
SC	548.059	555.918
-2 Log L	541.400	529.281

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	12.1190	3	0.0070
Score	13.1674	3	0.0043
Wald	12.4119	3	0.0061

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-2.0390	0.1660	150.8270	<.0001
smoke	1	0.7292	0.3198	5.1994	0.0226
sex	1	-0.5298	0.2853	3.4482	0.0633
smoke*sex	1	-0.0883	0.5471	0.0260	0.8718

The LOGISTIC Procedure

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	43.6	Somers' D	0.202
Percent Discordant	23.5	Gamma	0.301
Percent Tied	32.9	Tau-a	0.040
Pairs	59684	c	0.601